

Architecture Decision Record

Hybrid Multilingual Retrieval, Reranking, and Two-Pass Grounded Generation

Status: Accepted

Date: 2026-10-01

Project: Yemen Opportunity Navigator

Context

Yemen Opportunity Navigator must answer Arabic and English questions about opportunities using a curated source corpus while keeping answers grounded in source evidence. The current validated implementation contains 49 processed source records and 164 searchable chunks. The architecture must support semantic matching, exact-term matching, source attribution, measurable retrieval quality, and reproducible evaluation.

Decision

1. Cohere embed-multilingual-v3.0 for document and query embeddings.
2. Persistent Chroma for semantic vector retrieval.
3. BM25 with Arabic/English normalization for lexical retrieval.
4. Hybrid Vector + BM25 fusion using RRF to form a broader candidate set.
5. 20 hybrid candidates passed to Cohere rerank-multilingual-v3.0.
6. Top 5 reranked chunks used as the grounded generation context.
7. Command A (command-a-03-2025) in two passes: grounded draft, then draft-plus-context review.
8. Source attribution built separately from answer text, then rendered with the final answer.
9. Streamlit as the current UI and Supabase for authentication/session management.

Alternatives Considered

Vector-only retrieval was simpler and strong at 96.67% Recall@5, but missed one golden question.

BM25-only retrieval preserved exact lexical matching but reached only 80.00%.

Hybrid without reranking broadened candidate diversity but achieved 90.00%, below vector-only performance at the final top-five level.

Therefore, the accepted design keeps hybrid retrieval for candidate breadth and uses learned reranking to recover final precision and recall.

Supporting Evidence

docs/technical_evidence.md
docs/charts/architecture_diagram.png
evaluation/recall_summary.md
evaluation/ragas_summary.md
evaluation/investor_cost_summary.md
cost_analysis.md

Evidence - Retrieval

Configuration	Recall@5
Vector only	96.67% (29/30)
BM25 only	80.00% (24/30)
Hybrid + RRF	90.00% (27/30)
Hybrid + Cohere Reranker	100.00% (30/30)

The decisive improvement comes from reranking the broader hybrid candidate set, not from raw hybrid fusion alone. Final Recall@5 is also 100% for Arabic (15/15), English (15/15), and the hard subset (11/11).

Evidence - Generation Quality

RAGAS metric	Score
Faithfulness	0.9875
Answer Relevancy	0.8227
Context Precision	1.0000
Context Recall	1.0000

The project-level arithmetic mean across these four metric averages is 0.9526; it is a project summary, not a separate canonical RAGAS metric.

Operational Trade-off

The measured production path uses 2 Chat calls + 1 Embed call + 1 Rerank call per normal query. Across 30/30 successful cost-profile queries:

- Mean Command A generation cost: \$0.01982967/query
- P95 generation cost: \$0.02205750/query
- P50 end-to-end latency: 10.945 s
- P95 latency: 32.381 s

Consequences

Benefits: 100% Recall@5 on the current golden set, bilingual retrieval robustness, excellent context precision/recall, high faithfulness, and explicit source attribution.

Costs: extra API usage and latency from reranking and the second generation pass.

Known weakness: Answer Relevancy at 0.8227 is the primary measured quality improvement area.

Limitations: results are based on 30 retrieval questions, 20 RAGAS questions, and the current 49-source/164-chunk corpus. They are not guarantees for all future queries or larger-scale deployments.